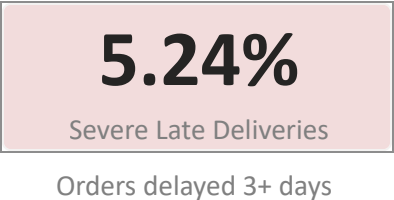
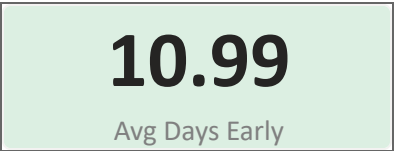
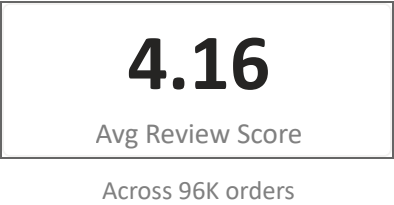


Olist Delivery Performance and Customer Satisfaction Analysis

Delivery performance can strongly influence customer satisfaction in e-commerce. Although most Olist orders are delivered on time, delayed deliveries may lead to lower review scores and poorer customer experiences. This analysis looks at how delivery timing and regional delivery trends relate to customer satisfaction.

Problem Statement: Analyze delivery performance by measuring delivery timeliness, identifying severe delays, and examining how delivery patterns affect customer review scores across regions. The goal is to determine which delivery-related issues are most connected to customer dissatisfaction and where operational improvements could have the greatest impact.

Brazilian E-Commerce Public Dataset by Olist (via Kaggle)



Executive Summary

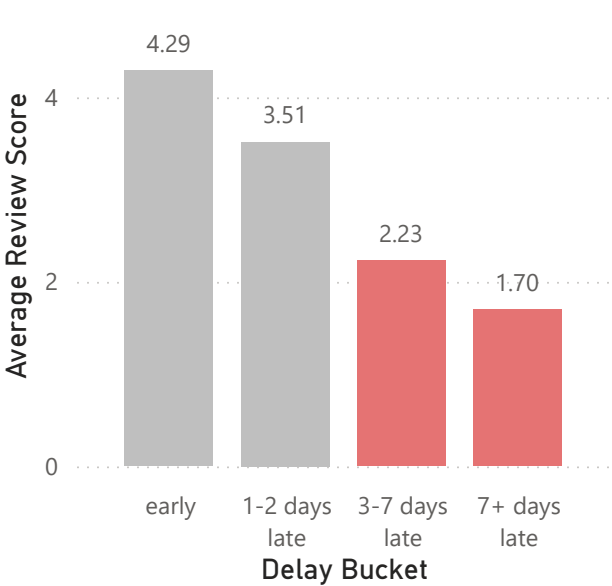
Key Factors of Customer Dissatisfaction:

- Severe delays (3+ days), associated with a sharp drop in average review score
- A small share (5.24%) of severely late orders linked to a 30% share of low reviews (score ≤ 2)
- The same high-volume product category experiencing varying standards of delivery performance across regions

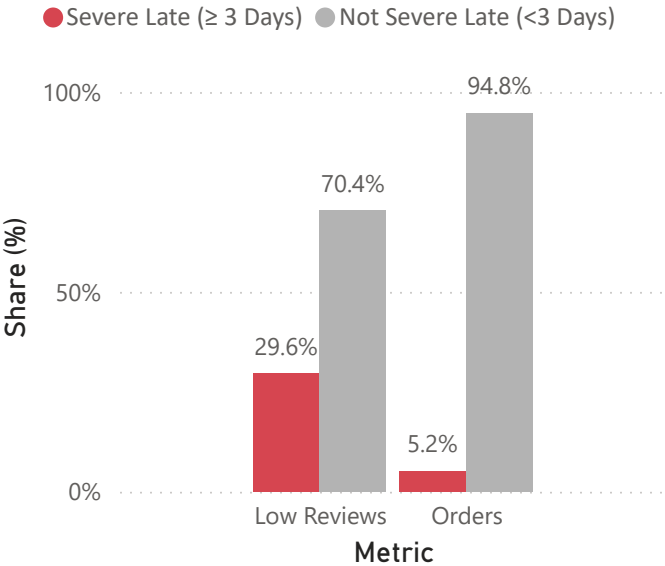
Business Recommendations:

- Use 3+ delay day threshold as an indicator for intervention
- Remediate 5.24% of severe late orders responsible for highest customer pain points, steering away from the less optimal ROI of the 94.8%
- Address differences in customer experience across regions, seeking to standardize best practices for delivery performance

Customer satisfaction drop 3+ days late



5% of orders linked to 30% of low reviews



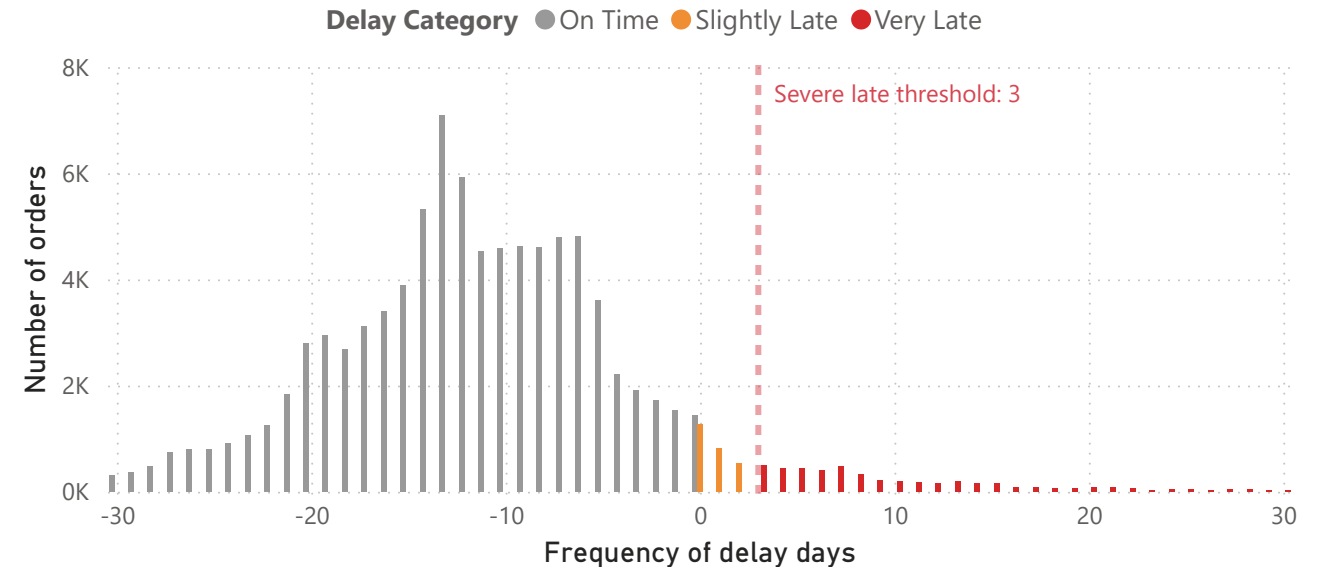
Same category, different outcomes

State	Total Orders	Severe Late %	Total Severe Late Orders	Average Review Score
RJ	1314	12.9%	170	3.79
SP	4229	3.0%	128	4.15
MG	1085	4.1%	45	3.93
RS	520	7.1%	37	3.99
BA	206	11.7%	24	3.81
SC	261	9.2%	24	3.96
ES	191	12.0%	23	3.85
GO	180	11.1%	20	3.69
DF	165	6.7%	11	3.94
CE	66	12.1%	8	4.05

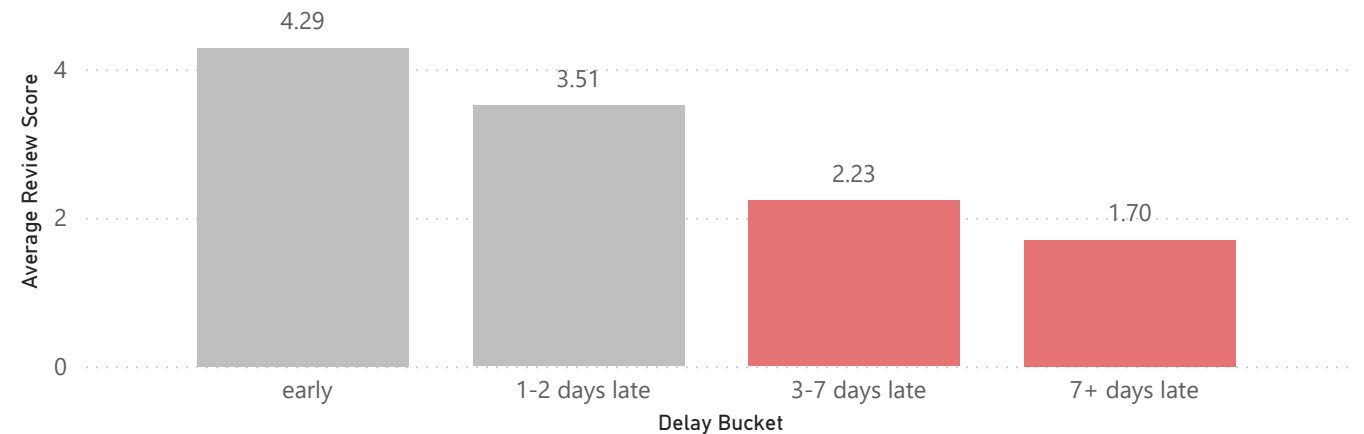
Key Factor 1: 3+ days late is linked to a severe drop in customer satisfaction

- Most orders in the Olist marketplace arrive early / on-time, with orders arriving on average ≈ 11 days early
- A small subset of orders (5.24%) that arrive 3+ days late, though uncommon, are associated with a 1.3 point drop in review score
- Logistics teams have a clear, 3+ day threshold to prevent collapsing customer satisfaction

Most Orders Arrive On Time, but a Small Late Tail Exists



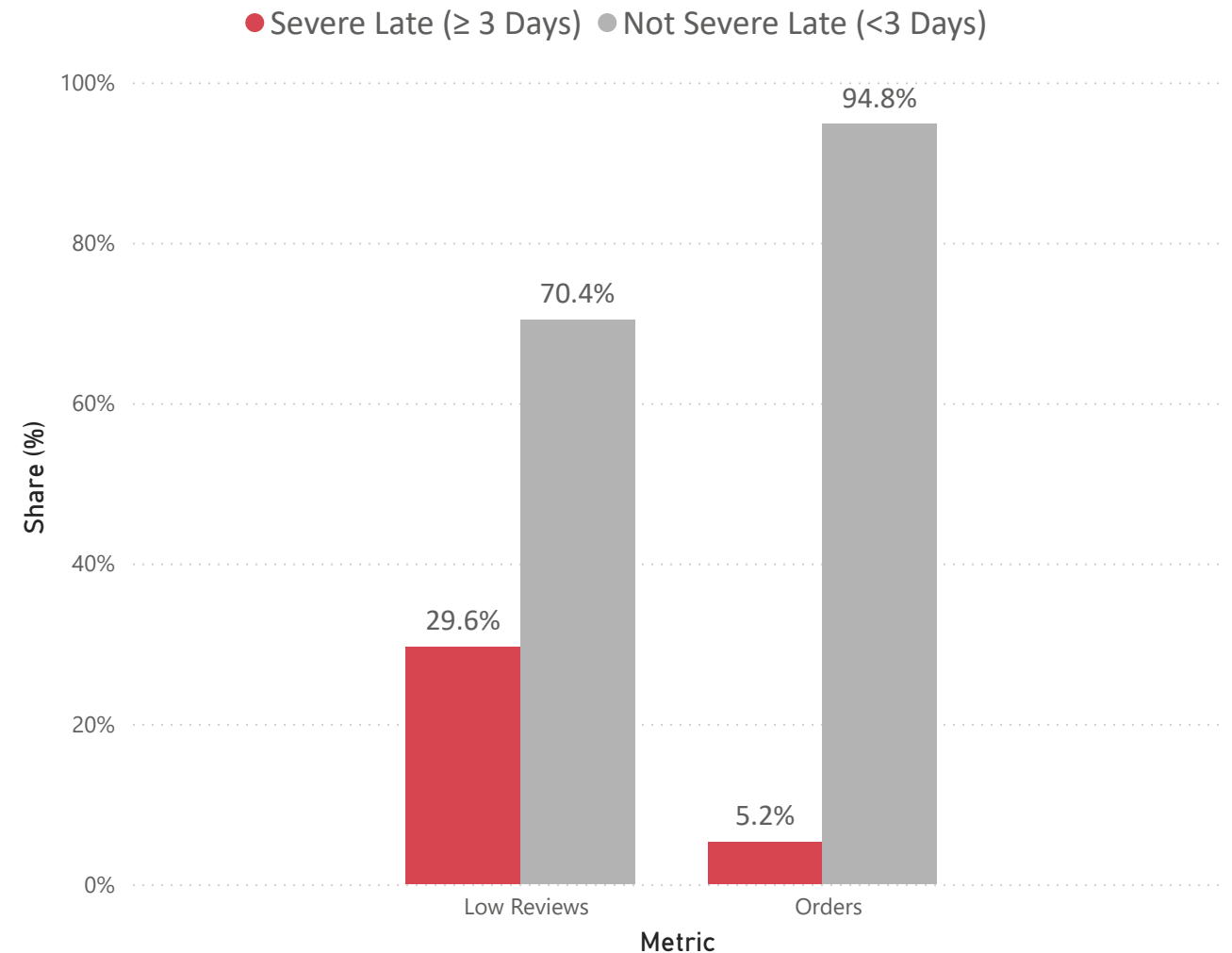
Customer satisfaction drops sharply after 3+ days late



Key Factor 2: 5.24% of orders are linked to a disproportionate share of customer dissatisfaction

- 5.24% of orders are severely late (3+ delay days)
- Severely late orders account for ≈30% of all low reviews (review score ≤ 2)
- Focusing on the small subset with high-impact operational failure yields a better ROI than improving the overall existing delivery standards

The same 5% of orders 3+ days late are linked to ≈30% of low reviews (score ≤ 2)



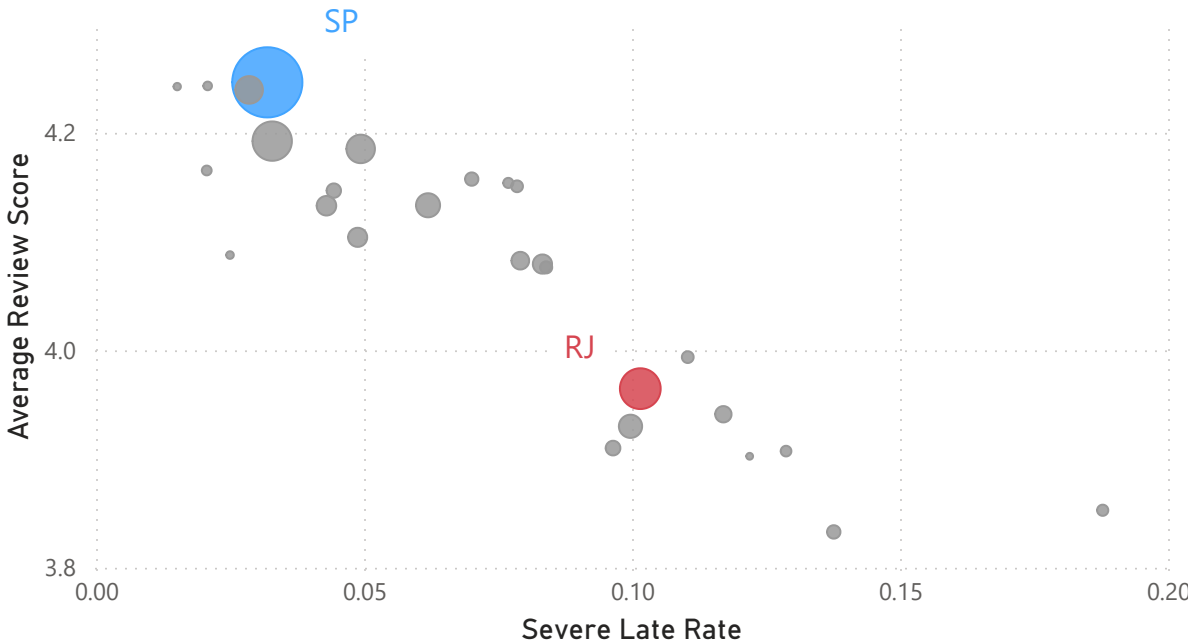
Key Factor 3: bed/bath/table product category experiences varying standards of delivery performance across different regions

- RJ and SP both have high-volume orders in the bed/bath/table category
- RJ experiences over 4x the severe late rate of SP (12.9% vs. 3.0%)
- Standardizing operational outcomes within the same category could address inconsistency in customer satisfaction from region to region

Same product category (bed/bath/table), different operational outcomes

State	Total Orders	Severe Late %	Total Severe Late Orders	Average Review Score
RJ	1314	12.9%	170	3.79
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RJ and SP: same in high-volume orders, same category, different in operational outcomes, RJ shows lower average satisfaction where severe-late rates are higher



Business Recommendations

Problem 	Recommendation 	Business Impact 
Orders 3+ late	Create early warning intervention system	Prevent steep review score decline
5.24% severe late orders are linked to 30% of low reviews	Prioritize CX remediation for severe late orders	Higher ROI operational focus
Regional inconsistency (RJ vs. SP)	Standardize regional logistics best practices	More consistent customer experience

Closing Insights

Final Takeaway: With averages like 4 for review score and 11 days early for deliveries, Olist does not have the problem of widespread delivery failure. Instead, the problem is concentrated operational failure within a small subset of severe late deliveries.

Core Findings

- 3+ day delays are associated with a sharp drop in satisfaction
- ≈5% of orders are linked to 30% of low reviews
- Delivery performance varies regionally

Business Implications

- Broad operational overhaul is unnecessary
- The highest ROI comes from targeting severe late deliveries
- Regional standardization may improve CX consistency

